CodeSport

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As per our readers' request, this month's column is a 'New Year' special featuring more programming-related interview questions.

Many readers have sent the answers to the questions we featured in our Diwali issue. Some requested that we keep open the challenging questions till the readers come up with the solutions themselves. Hence we will not discuss the solutions to the remaining questions in this column but keep it open till next month. Congrats to our readers Mani Ravichandran, Wallace Jacob and Nilesh Kumar for getting most of the answers correct. Please do keep your solutions and feedback coming in.

Many readers had also mailed me requesting for more programming-related interview questions. So here's a 'New Year Special' of CodeSport to challenge you over this holiday season. Since readers had asked for questions from the programming languages perspective, we've included a few questions on the C/C++ language.

This month's questions

1. In a given undirected graph G (V, E), V represents the set of vertices and E represents the set of edges; and a weight function c(u,v) associates a non-negative weight with the edges of the graph. A minimum weighted cycle in the graph is defined as the cycle whose sum of edge weights is minimum over all cycles in the graph. A maximum weighted cycle is one whose sum of edge weights is maximum over all the cycles in the graph.
    a. Can you come up with an algorithm to find the minimum weighted cycle? What is the complexity of your algorithm?
    b. Can you come up with an algorithm to find the maximum weighted cycle? What is the complexity of your algorithm?
    c. Can you come up with a polynomial time algorithm for this problem? If not, why?
2. In a given array of N unsorted integers, you are asked to find the smallest 20 integers from that array. Can you come up with an algorithm with a complexity of O (N)? *Hint: Remember the algorithm to find the linear time median.*
3. In a given array of N unsorted integers, your friend keeps asking you to retrieve the minimum element, remove it from the array and print it. He keeps doing so until 'k' elements are retrieved. However, he does not tell you the value of 'k' ahead of time. So you cannot use the solution from question (2) above. He can stop asking for the minimum element at any time, and hence the value of 'k' is determined dynamically. In such a scenario,
    a. Can you come up with an algorithm for this problem that has a complexity of O(Nk)?
    b. Can you come up with an algorithm which has a complexity of O(N) + klogN?
4. Here is a question that is often asked in Google programming interviews. You are given an array of N integers. You have to find the sub-array out of this array whose sum is the maximum, with the added condition that no two elements of the sub-array are neighbours in the original array. Can you come up with a polynomial time algorithm for solving this problem?
5. In most programming interviews you are expected to be quite comfortable with various graph algorithms. In particular, Single Source Shortest Path Algorithms are favourites with many interviewers. These algorithms vary in the number of edge relaxations that are needed to obtain the SSSP solution. Djikstra's single source shortest path (SSSP) algorithm can be used when the graph contains no negative weighted edges. Given a graph that can contain negative weighted edges, but no negative weighted cycles, can you modify Djikstra's algorithm to come up with the SSSP without impacting the complexity of Djikstra's algorithm?
6. You have an array of N elements and you have to find the sub array out of this array whose sum is the maximum. The condition is that no two elements that you choose in your sub array should be adjacent to each other in the original array. How will you do this polynomial time (incomplete?)?
7. Do you know what the term RAIL (Resource Acquisition Is Initialisation) means? Under